

XGBoost Hyperparameter Dials – Chosen Values, Ranges, and Notes

Shaded arc = commonly effective region; needle = your current choice



[0.01 – 0.3]

Smaller → slower, safer learning



[50 – 3000]

Capacity .; use early stopping



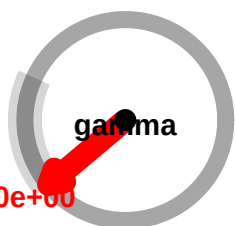
[2 – 12]

Controls interactions/non-linearity



[1 – 20]

Higher → smoother, fewer tiny leaves



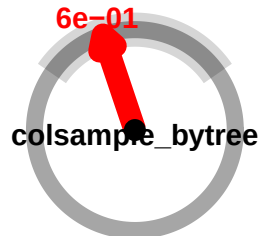
[0 – 20]

Higher → prunes weak splits



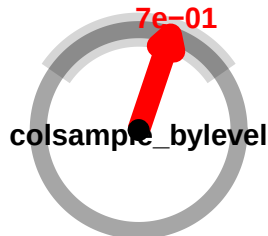
[0.5 – 1]

Row sampling per tree; regularizes



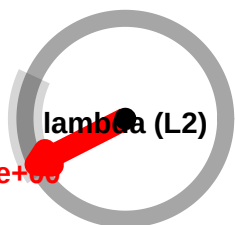
[0.3 – 1]

Feature sampling per tree



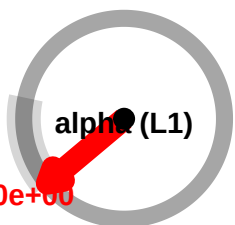
[0.3 – 1]

Feature sampling per split level



[0 – 20]

Weight shrinkage; stabilizes



[0 – 10]

Sparsity; can aid calibration



[0 – 10]

Stabilizes rare-event logits



[0.1 – 20]

Handle class imbalance (. n_neg/n_pos)